

Code-Mixed Gateway: Cascade Routing — Status Report

Observations, data and measurements as of 28 Sept 2026. No recommendations in this document.

1. Goal

Route each Hinglish user prompt to a cheap local model (Qwen 2.5 7B, free) when it can handle it, else to a premium model (Gemini, paid). A prompt may go cheap only if the failure rate among cheap-routed prompts is certifiably at most 5% ($\alpha = 0.05$).

2. Data inventory

File	Rows	Contents
prompts.csv	10,273 unique (10,378 with dupes)	Source prompts, single column
prompts_scored.csv	10,378	m9 difficulty d + budget features per prompt
prompts_routed.csv	10,378	Original Laya router labels (Yes 8,035 / No 2,343)
prompts_routed_v2.csv	10,378	Fine-tuned router labels (Yes 6762 / No 3616)
judge_output_200.csv (v1)	200	Claude grades Y:123 / N:77
judge_output_200_v2.csv (v2)	200	Claude re-grades Y:119 / N:81; 21 rows use uncapped Qwen answers
qwen_redo_uncapped.csv	21	Qwen answers regenerated with no token cap
calibration_v2.csv	200 grid points	m9-d grid ($\tau^* = \text{None}$)
scored_laya_10378.csv	10,273 unique	Laya P(No) difficulty per prompt
calibration_laya.csv	200 grid points	P(No) grid ($\tau^* = 0.5528$)
pilot_300.csv / qwen_pilot_300.csv	300	Stratified pilot prompts + Colab Qwen answers
judge_output_300.csv	300	Claude pilot grades Y:215 / N:85
prompts_rerouted_200.csv	200	Fine-tuned router re-run on first 200

3. Difficulty scores (m9 reasoning budget, normalized)

Each prompt's thinking-token budget ($128 + 120 \cdot \text{codemix} + 96 \cdot \text{math} + 48 \cdot \text{logic} + 8 \cdot \min(\text{len}, 40)$), capped at 2048) is mapped to $d = (\text{budget} - 128) / (2048 - 128)$. Observed: min 0.0167, mean 0.1097, max 0.6224. No scores pile at 0 or 1.

4. Router labels: before vs after fine-tuning

Router	Yes (cheap)	No (premium)
Original (zero-shot)	8,035 (77%)	2,343 (23%)
Fine-tuned (Colab T4, 140 train rows)	6762 (65%)	3616 (35%)

5. Empirical grades (Claude, Qwen vs premium answers)

200 prompts graded Y (Qwen adequate) / N (not). v1: Y:123/N:77. v2 re-grade: Y:119/N:81 — 22 labels changed (9 N→Y, 13 Y→N). One premium row observed misaligned (omelette recipe filed under a bus-story prompt).

6. Router-vs-reality agreement (200 graded rows)

Router	Agreement with Claude	Fail rate among router-Yes
Original	114/200 (57.0%)	72/181 (39.8%)
Fine-tuned	181/200 (90.5%)	14/128 (10.9%)

Of the 90 v2 disagreements, 30 were router judgments at over 80% confidence. Mean router confidence: 0.73 on agreements vs 0.67 on disagreements — confidence barely separates right from wrong.

7. Uncapped re-run (21 truncation suspects)

21 N-graded answers longer than 1200 chars were regenerated with no token cap (num_predict = -1, run-until-EOS). Zero runaways; 19/21 ended on clean terminal punctuation. Re-graded: 7 flipped N→Y (cap was the cause), 14 stayed N (fail on the merits), 0 flipped Y→N.

8. Difficulty-vs-outcome separation

Mean m9 d: Claude-Y 0.1180 (n=119) vs Claude-N 0.1103 (n=81). The score distributions overlap almost completely; difficulty as computed does not separate pass from fail.

9. Conformal calibration (200-point grid, alpha 0.05, delta 0.05)

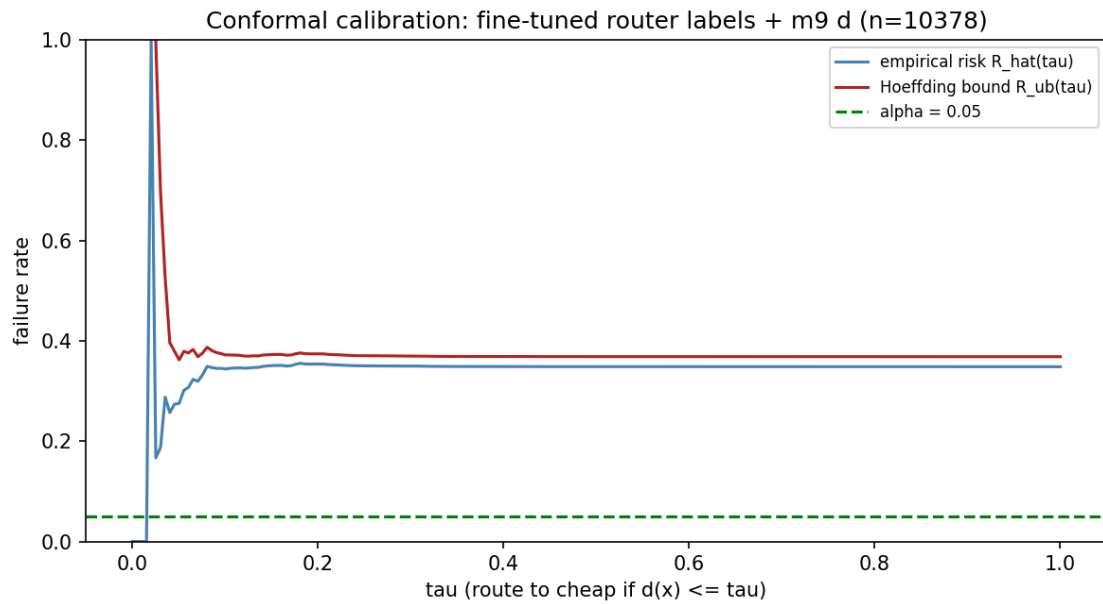
Method: for each tau, failure rate among prompts with $d \leq \tau$, plus Hoeffding margin $\sqrt{\log(200/0.05)/(2n)}$. Join coverage: 10,378/10,378, zero missing. Base Yes rate: 65.2%.

tau	n	R_hat	R_ub
0.0	0	0.0	inf
0.005	0	0.0	inf
0.0101	0	0.0	inf
0.0151	0	0.0	inf
0.0201	1	1.0	3.0364
0.0251	6	0.1667	0.998
0.0302	16	0.1875	0.6966
0.0352	73	0.2877	0.526
0.0402	214	0.257	0.3962
0.0452	369	0.2737	0.3797

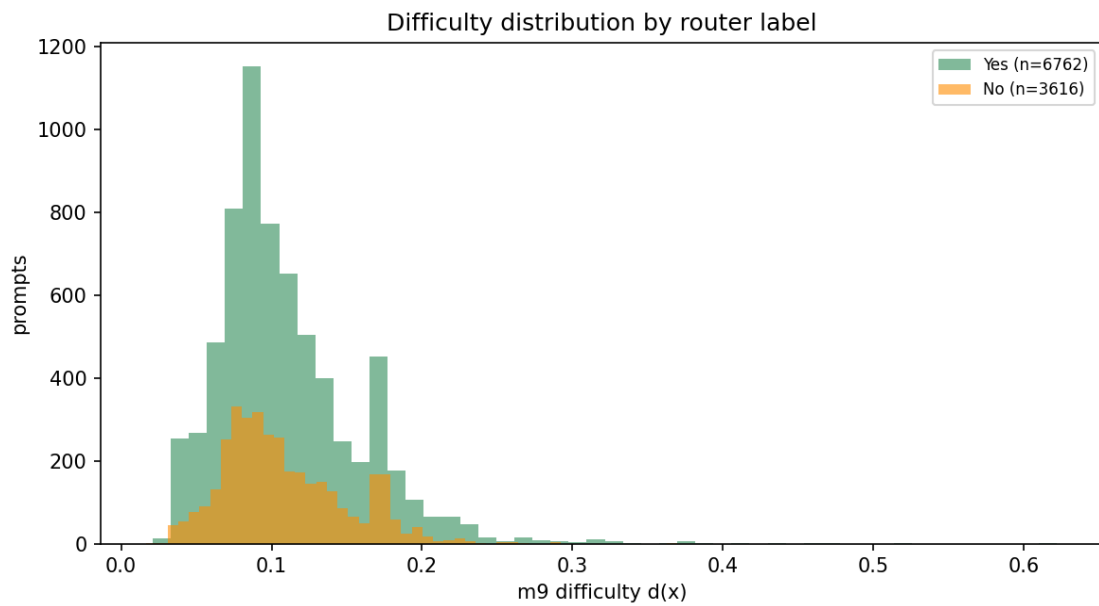
Result: tau* = None — no grid point satisfies $R_{ub} \leq 0.05$. Lowest empirical risk with $n > 100$ is 0.257 (tau $\approx$ 0.0402, n = 214). The single easiest prompt in the dataset (d = 0.0167, 'Healthcare marketing strategy banao') is itself labeled No.

Worked margin example (tau = 0.05, n = 552): $\sqrt{\log(200/0.05)/(2 \times 552)} \approx 0.087$, i.e. ± 8.7 points of statistical doubt.

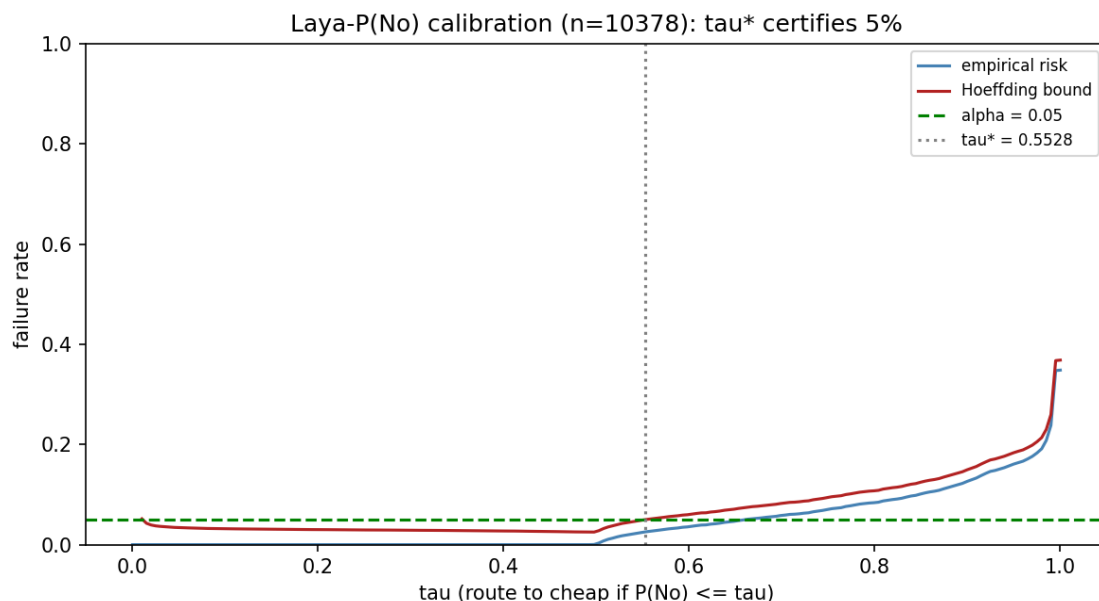
10. Figures



m9-difficulty risk curve. The bound never reaches the 0.05 line ($\tau^* = \text{None}$).



m9 difficulty distribution split by fine-tuned router label.



Laya-P(No) risk curve. Bound crosses 0.05 at $\tau^* = 0.5528$ (coverage 66.9%).

11. Difficulty redefined: Laya P(No)

Motivation: m9 d means were 0.1180 (pass) vs 0.1103 (fail) — no separation. $d(x)$ was redefined as the fine-tuned router's own $P(\text{No})$. Validation on the 200: Claude-Y mean $d = 0.082$ ($n=119$) vs Claude-N mean $d = 0.843$ ($n=81$). Full set scored: 10273 unique prompts, d mean 0.414, range 0.005–0.995. Determinism check (200 prompts scored in two separate runs): max difference 0.0001. Circularity note: the same model output supplies both score ($P(\text{No})$) and label (argmax); the calibration below therefore certifies score-label consistency, with ranking power independently confirmed by Claude.

12. Conformal calibration on Laya P(No) (200-point grid, alpha 0.05, delta 0.05)

Join: 10,378/10,378 v2 labels matched to $P(\text{No})$ scores, zero missing. Base Yes rate 65.2%. Result: $\tau^* = 0.5528$ — the largest τ with bound ≤ 0.05 (all higher taus fail; verified by independent recompute). Coverage 66.9% (6939 cheap), empirical risk 0.0255. Empirical spot-check on the 200 Claude rows at τ^* : 128 routed cheap, 14 fail (10.9%; $n=128$, wide margins).

13. Stratified 300 pilot (fresh prompts, 30 per P(No) decile)

Qwen answers generated on Colab T4 via Ollama (same quant/cap); mid-run disconnect recovered through resume; file verified 300/300 in order, zero empty. GPT premium answers supplied; Claude grades: Y:215 / N:85.

Decile	n	Fail rate
0	30	0.33
1	30	0.30
2	30	0.17
3	30	0.40

Decile	n	Fail rate
4	30	0.23
5	30	0.30
6	30	0.30
7	30	0.20
8	30	0.17
9	30	0.43

Headline: agreement with fine-tuned router 148/300 (49.3%). Router-Yes fail rate 29.1%. Fail rate shows no rise with difficulty decile — ranking power observed on the training pool does not transfer to fresh prompts.

14. Model and run versions

Item	Value
Cheap model	qwen2.5:latest (7.6B Q4, local Ollama CPU ~6 tok/s; Colab T4 for pilot, same quant)
Gemini API role	gemini-3.8-flash: connectivity probes + abandoned judge attempts only; never graded
Premium baseline answers (200 + 300)	GPT (user-supplied; judge rubric is model-agnostic)
Empirical grader	Claude (200: v1 + v2 rounds; 300 pilot: one round)
Router base	Laya / ModernBERT-large 395M; fine-tuned on Colab T4, 140 rows, 12 epochs
Router checkpoint	D:/laya, model_name=laya-hinglish-cheap, fine_tuned=true; choice-temp fit hit 10.0 clamp (loader clamps to 5, confidences uncalibrated)
Rollback copy	None — Colab files overwrote D:/laya in place; pre-finetune zip at D:/Programming/laya_ckpt.zip (verify date before trusting)